

Chapter 5 Group Work

SOLUTIONS

Problem 1

Suppose that you randomly select a student from a large population of 4th grade students and record the student's sex and number of siblings. Let $M = 1$ if the selected student is male and $M = 0$ if the selected student is female. Let S record the student's number of siblings. The proportion of the population of students falling into each category of M and S is recorded in the table below.

# of siblings:	0	1	2	3	4
Female	.10	.18	.12	.07	.04
Male	.12	.18	.14	.03	.02

- Verify that the above table is a valid joint pmf.
- Find the probability that the randomly selected student will be male with two siblings
- Find $P(M = 0, S \leq 2)$
- Find the probability that the randomly selected student will be an only child
- Find $F(1, 1)$

Solution

Part a

$$0.10 + 0.12 + 0.18 + 0.18 + 0.12 + 0.14 + 0.07 + 0.03 + 0.04 + 0.02 = 1$$

Part b

$$P(M = 1, S = 2) = 0.14$$

Part c

$$P(M = 0, S \leq 2) = 0.12 + 0.18 + 0.14 = 0.44$$

Part d

$$P(S = 0) = 0.10 + 0.12 = 0.22$$

Part e

$$F(1, 1) = P(M \leq 1, S \leq 1) = P(S \leq 1) = 0.10 + 0.12 + 0.18 + 0.18 = 0.58$$

Problem 2

Let Y_1 and Y_2 have the joint pdf

$$f(y_1, y_2) = \begin{cases} e^{-(y_1+y_2)}, & y_1 > 0, y_2 > 0 \\ 0, & \text{elsewhere} \end{cases}$$

a. What is $P(Y_1 < 1, Y_2 > 5)$? b. What is $P(Y_1 + Y_2 < 3)$?

Part a

$$\begin{aligned} P(Y_1 < 1, Y_2 > 5) &= \int_0^1 \int_5^\infty e^{-(y_1+y_2)} dy_2 dy_1 \\ &= \int_0^1 \left(-e^{-(y_1+y_2)} \Big|_{y_2=5}^\infty \right) dy_1 \\ &= \int_0^1 e^{-(y_1+5)} dy_1 \\ &= -e^{-(1+5)} + e^{-5} \\ &= .00426. \end{aligned}$$

Part b

$$\begin{aligned} P(Y_1 + Y_2 < 3) &= P(Y_1 < 3 - Y_2) \\ &= \int_0^3 \left(\int_0^{3-y_1} e^{-(y_1+y_2)} dy_2 \right) dy_1 \\ &= 1 - 4e^{-3} \\ &= .8009. \end{aligned}$$

Problem 3

Return to the scenario in Problem 1

- Find the marginal distribution of M
- Find the marginal distribution of S
- Find $P(S \leq 2 | M = 1)$
- Find $P(M = 1 | S \leq 2)$

Solutions

Part a

	P(M = m)
Female (M = 0)	0.51
Male (M = 1)	0.49

Part b

# of siblings (S):	0	1	2	3	4
P(S = s)	.22	.36	.26	.10	.06

Part c

$$P(S \leq 2 \mid M = 1) = \frac{P(S \leq 2, M = 1)}{P(M = 1)}$$

- $P(M = 1) = 0.49$
- $P(S \leq 2, M = 1) = P(S = 0, M = 1) + P(S = 1, M = 1) + P(S = 2, M = 1) = 0.12 + 0.18 + 0.14 = 0.44$

$$P(S \leq 2 \mid M = 1) = \frac{0.44}{0.49} \approx 0.898$$

Part d

$$P(M = 1 \mid S \leq 2) = \frac{P(S \leq 2, M = 1)}{P(S \leq 2)}$$

- $P(S \leq 2) = P(S = 0) + P(S = 1) + P(S = 2) = 0.22 + 0.36 + 0.26 = 0.84$
- $P(S \leq 2, M = 1) = 0.44$ (from above)

$$P(M = 1 \mid S \leq 2) = \frac{0.44}{0.84} \approx 0.524$$

Problem 4

Let $f(x, y) = \frac{4}{3}(1 - xy)$, $0 \leq x \leq 1$, $0 \leq y \leq 1$.

- Find the marginal pdfs of X and Y .
- Find $P(X \leq Y/2)$

$$f(x, y) = \frac{4}{3}(1 - xy), \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

$$f_X(x) = \int_0^1 f(x, y) dy = \int_0^1 \frac{4}{3}(1 - xy) dy = \frac{4}{3} \left[y - \frac{1}{2}xy^2 \right]_0^1 = \frac{4}{3} \left(1 - \frac{1}{2}x \right) = \frac{4}{3} - \frac{2}{3}x, \quad 0 < x < 1$$

$$f_Y(y) = \int_0^1 f(x, y) dx = \int_0^1 \frac{4}{3}(1 - xy) dx = \frac{4}{3} \left[x - \frac{1}{2}yx^2 \right]_0^1 = \frac{4}{3} \left(1 - \frac{1}{2}y \right) = \frac{4}{3} - \frac{2}{3}y, \quad 0 < y < 1$$

$$P(X \leq Y/2) = \int_0^1 \int_0^{y/2} f(x, y) dx dy = \int_0^1 \int_0^{y/2} \frac{4}{3}(1 - xy) dx dy = \frac{7}{24}$$

Problem 5

Let the joint pmf of X and Y be

$$f(x, y) = \frac{xy^2}{30}, \quad x = 1, 2, 3, \quad y = 1, 2$$

- What are the marginal distributions of X and Y ?
- Are X and Y independent?
- What is $P(X > 1)$?
- What is $E(X)$?

Part a

$$f_X(x) = \sum_{y=1}^2 \frac{xy^2}{30} = \frac{x \cdot 1^2}{30} + \frac{x \cdot 2^2}{30} = \frac{x}{30} + \frac{4x}{30} = \frac{5x}{30} = \frac{x}{6}, \quad x = 1, 2, 3$$

$$f_Y(y) = \sum_{x=1}^3 \frac{xy^2}{30} = \frac{1 \cdot y^2}{30} + \frac{2 \cdot y^2}{30} + \frac{3 \cdot y^2}{30} = \frac{6y^2}{30} = \frac{y^2}{5}, \quad y = 1, 2$$

Part b

Since $f_X(x)f_Y(y) = \frac{x}{6} \cdot \frac{y^2}{5} = \frac{xy^2}{30} = f(x, y)$ for all x, y , X and Y are independent.

Part c

$$P(X > 1) = f_X(2) + f_X(3) = \frac{2}{6} + \frac{3}{6} = \frac{5}{6}$$

Part d

$$E(X) = \sum_{x=1}^3 xf_X(x) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{2}{6} + 3 \cdot \frac{3}{6} = \frac{14}{6} = \frac{7}{3}$$

Problem 6

Let X and Y be two continuous random variables with joint pdf $f(x, y) = \frac{3}{16}xy^2$, $0 \leq x \leq 2$, $0 \leq y \leq 2$. Are the two random variables independent?

Solution

Both are bounded between constants, and we can define $g(x) = \frac{3}{16}x$ and $h(y) = y^2$, so by Theorem 5.5 they are independent.

Problem 7

Let X and Y have the joint pdf $f(x, y) = x + y$, $0 \leq x \leq 1$, $0 \leq y \leq 1$. Find the marginal pdfs $f_X(x)$ and $f_Y(y)$ and show that X and Y are dependent.

Solution

$$f_x(x) = \int_0^1 (x+y)dy = x + \frac{1}{2}, \quad 0 \leq x \leq 1$$

$$f_y(y) = \int_0^1 (x+y)dx = y + \frac{1}{2}, \quad 0 \leq y \leq 1$$

$$f_x(x)f_y(y) = (x + \frac{1}{2})(y + \frac{1}{2}) \neq x + y$$

Problem 8

Let Y_1, Y_2, Y_3 be independent random variables with binomial distributions $b(4, 1/2)$, $b(6, 1/3)$, and $b(12, 1/6)$.

- Find $E(Y_1 Y_2 Y_3)$
- Find the mean and variance of $W = Y_1 + Y_2 + Y_3$.

Part a

Due to independence, $E(Y_1 Y_2 Y_3) = E(Y_1)E(Y_2)E(Y_3)$. By binomial distribution, $E(Y_i) = n_i p_i$, so $E(Y_1 Y_2 Y_3) = 4(1/2) * 6(1/3) * 12(1/6) = 2 * 2 * 2 = 8$.

Part b

$E(W) = E(Y_1 + Y_2 + Y_3) = E(Y_1) + E(Y_2) + E(Y_3) = 6$ By independence, $V(Y_1 + Y_2 + Y_3) = V(Y_1) + V(Y_2) + V(Y_3) = 4(1/2)(1/2) + 6(1/3)(2/3) + 12(1/6)(5/6) = 4$

Problem 9

Show that $Cov(X, Y) = E(XY) - \mu_X \mu_Y$.

Hint: start with the definition $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

Solution

$$\begin{aligned} Cov(X, Y) &= E((X - \mu_X)(Y - \mu_Y)) \\ &= E(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\ &= E(XY) - \mu_Y E(X) - \mu_X E(Y) + \mu_X\mu_Y \\ &= E(XY) - \mu_Y\mu_X - \mu_X\mu_Y + \mu_X\mu_Y \\ &= E(XY) - \mu_X\mu_Y \end{aligned}$$

Problem 10

Show that $E(XY) = \mu_X\mu_Y + \rho_{XY}\sigma_X\sigma_Y$

Solution

By re-arranging the identity proven in Problem 1, we have

$$E(XY) = Cov(X, Y) + E(X)E(Y)$$

Since the correlation is defined as $\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$, we have $Cov(X, Y) = \rho_{XY} \sigma_X \sigma_Y$. Plugging in gives

$$E(XY) = \rho_{XY} \sigma_X \sigma_Y + E(X)E(Y)$$

Problem 11

Show that if two random variables X and Y are independent, then their covariance is 0.

Hint: start with the shortcut formula $Cov(X, Y) = E(XY) - \mu_X \mu_Y$ and re-write $E(XY)$ as a double sum.

Solution

Let X, Y be independent. We will therefore use the fact that $f(x, y) = f_x(x)f_y(y)$

$$\begin{aligned} Cov(X, Y) &= E((X - \mu_X)(Y - \mu_Y)) \\ &= \sum_x \sum_y xyf(x, y) - \mu_X \mu_Y \\ &= \sum_x \sum_y xyf_x(x)f_y(y) - \mu_X \mu_Y \\ &= \sum_x xf_x(x) \sum_y yf_y(y) - \mu_X \mu_Y \\ &= E(X)E(Y) - \mu_X \mu_Y \\ &= 0 \end{aligned}$$

Problem 12

Return to the scenario in Problem 1.

- Are M and S independent?
- Find $E(M)$ and $V(M)$
- Find $E(S)$ and $V(S)$
- Find $Cov(M, S)$
- Suppose that instead of just one student, you sample 12 students. Let $S_1, \dots, S_{12}$ represent the number of siblings reported by each student (you can assume that these random variables are independent). Let $\bar{S} = \sum_{i=1}^{12} \frac{S_i}{12}$ be the average number of siblings in the sample. Find $E(\bar{S})$ and $V(\bar{S})$

Part a

Independent if and only if $P(S = s)P(M = m) = P(S = s, M = m)$ for all s, m . For the coordinate (0,1), we have $P(S = 0)P(M = 0) = 0.22 * 0.51 = 0.1122 \neq P(S = 0, M = 0) = .10$, therefore S and M are NOT independent.

$$\sigma_y^2 = \frac{11}{144}$$

Part b

Using the marginal distribution found above, we have

$$\begin{aligned} E(M) &= 0(.51) + 1(.49) = 0.49 \\ E(M^2) &= 0^2(.51) + 1^2(.49) = 0.49 \\ V(M) &= 0.49 - (0.49)^2 = 0.2499 \end{aligned}$$

Part c

Using the marginal distribution found above, we have

$$\begin{aligned} E(S) &= 0(.22) + 1(.36) + 2(.26) + 3(.10) + 4(.06) = 1.42 \\ E(S^2) &= 0^2(.22) + 1^2(.36) + 2^2(.26) + 3^2(.10) + 4^2(.06) = 3.26 \\ V(S) &= 3.26 - (1.42)^2 = 1.2436 \end{aligned}$$

Part d

$$\begin{aligned} Cov(M, S) &= \sum_m \sum_s msf(m, s) \\ &= (1)(1)(.18) + (1)(2)(.14) + (1)(3)(.03) + (1)(4)(.02) \\ &= 0.63 \end{aligned}$$

Note that the other 6 (m,s) coordinate pairs have at least one 0, so the term cancels out of double sum

Part e

$$\begin{aligned} E(\bar{S}) &= E\left(\sum_{i=1}^{12} \frac{S_i}{12}\right) \\ &= \sum_{i=1}^{12} E\left(\frac{S_i}{12}\right) \quad \text{b/c } E() \text{ is a linear operator} \\ &= \sum_{i=1}^{12} \frac{1}{12} E(S_i) \\ &= \sum_{i=1}^{12} \frac{1}{12} (1.42) \\ &= \frac{1}{12} (1.42) \sum_{i=1}^{12} 1 \\ &= \frac{1}{12} (1.42) (12) \\ &= 1.42 = E(S_i) \end{aligned}$$

Problem 13

Show that for any constants a and b , $Cov(a + X, b + Y) = Cov(X, Y)$. That is, shifting by a constant does not change the covariance. *Note: this fact will be useful on HW 5.110*

SOLUTION

By definition,

$$Cov(a + X, b + Y) = E[(a + X)(b + Y)] - E[a + X]E[b + Y].$$

Compute:

$$E[(a + X)(b + Y)] = ab + aE[Y] + bE[X] + E[XY],$$

and

$$E[a + X]E[b + Y] = ab + aE[Y] + bE[X] + E[X]E[Y].$$

Thus,

$$Cov(a + X, b + Y) = E[XY] - E[X]E[Y] = Cov(X, Y).$$

Problem 14

To estimate a proportion of units that meet a given criteria, we often use the estimator $\hat{p} = \frac{Y}{n}$, where $Y \sim \text{binomial}(n, p)$. Find the expected value and variance of $\hat{p}$, assuming n is fixed.

SOLUTION

Let

$$\hat{p} = \frac{Y}{n}, \quad Y \sim \text{Binomial}(n, p),$$

with n fixed.

Since $E[Y] = np$,

$$E[\hat{p}] = E\left[\frac{Y}{n}\right] = \frac{1}{n}E[Y] = \frac{np}{n} = p.$$

Since $V(Y) = np(1-p)$,

$$V(\hat{p}) = V\left(\frac{Y}{n}\right) = \frac{1}{n^2}V(Y) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

Problem 15

A learning experiment requires a rat to run a maze (a network of pathways) until it locates one of three possible exits. Exit 1 presents a reward of food, but exits 2 and 3 do not. (If the rat eventually selects exit 1 almost every time, learning may have taken place). Let Y_i denote the number of times exit i is chosen in successive runnings. For the following, assume that the rat chooses an exit at random on each run.

- Find the probability that $n = 6$ runs result in $Y_1 = 3$, $Y_2 = 1$ and $Y_3 = 2$.
- For general n , find $E(Y_1)$ and $V(Y_1)$.
- For general n , find $Cov(Y_2, Y_3)$.
- To check for the rat's preference between exits 2 and 3, we may look at $Y_2 - Y_3$. Find $E(Y_2 - Y_3)$ and $V(Y_2 - Y_3)$ for general n .

SOLUTION

- a. Using the multinomial distribution with $p_1 = p_2 = p_3 = \frac{1}{3}$,

$$P(Y_1 = 3, Y_2 = 1, Y_3 = 2) = \frac{6!}{3!1!2!} \left(\frac{1}{3}\right)^6 = 0.0823.$$

b.

$$E(Y_1) = \frac{n}{3}, \quad V(Y_1) = n \left(\frac{1}{3}\right) \left(\frac{2}{3}\right) = \frac{2n}{9}.$$

c.

$$Cov(Y_2, Y_3) = -n \left(\frac{1}{3}\right) \left(\frac{1}{3}\right) = -\frac{n}{9}.$$

d.

$$E(Y_2 - Y_3) = \frac{n}{3} - \frac{n}{3} = 0,$$

$$V(Y_2 - Y_3) = V(Y_2) + V(Y_3) - 2Cov(Y_2, Y_3) = \frac{2n}{3}.$$

Problem 16

The number of defects per yard in a certain fabric, Y , is known to have a Poisson distribution with parameter λ . The parameter λ is assumed to be a random variable with a pdf given by

$$f(\lambda) = e^{-\lambda}, \quad \lambda \geq 0$$

- Find the expected number of defects per yard by first finding the conditional expectation of Y for given λ .
- Find the variance of Y .

SOLUTION

Let

$$Y \mid \lambda \sim \text{Poisson}(\lambda), \quad f(\lambda) = e^{-\lambda}, \lambda \geq 0.$$

a. Expected value

The conditional expectation is $E(Y \mid \lambda) = \lambda$. by properties of Poisson distribution. Thus,

$$E(Y) = E[E(Y \mid \lambda)] = E(\lambda).$$

Since $\lambda \sim \text{Exponential}(1)$,

$$E(\lambda) = \int_0^\infty \lambda e^{-\lambda} d\lambda = 1.$$

Therefore,

$$E(Y) = 1.$$

b. Variance

Using the law of total variance,

$$V(Y) = E[V(Y \mid \lambda)] + V(E(Y \mid \lambda)).$$

For a Poisson random variable,

$$V(Y \mid \lambda) = \lambda, \quad E(Y \mid \lambda) = \lambda.$$

Hence,

$$V(Y) = E(\lambda) + V(\lambda).$$

For $\lambda \sim \text{Exponential}(1)$,

$$E(\lambda) = 1, \quad V(\lambda) = 1.$$

Therefore,

$$V(Y) = 1 + 1 = 2.$$